Classification Models
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1 Introduction

This paper investigates machine learning classification techniques to classify handwritten digits from the MNIST data set. A range of methods are applied and compared on accuracy, macro score, and training time then evaluated to figure out the best computational method.

2 Data Exploration

The MNIST dataset that is used in this paper which contains 70,000 handwritten digits ranging from 0-9. There are 784 features for each digit, these represent a pixel in the 28x28 grayscale image MNIST uses. Every digit contains 5000-7000 images with 1 containing the most (6724) and 5 the least (5421).Examples of these are shown in figure 1.

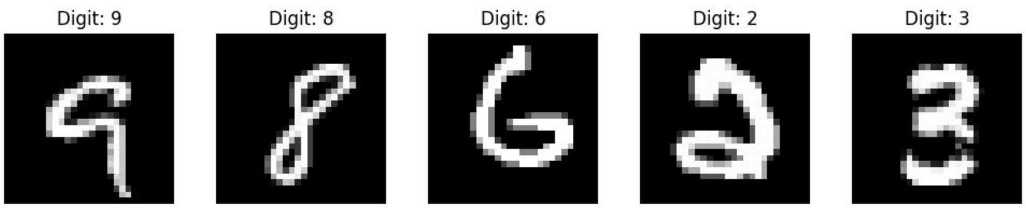

Figure 1: Example Digits

An 80/20 Train/Test split is applied to the images where the train set has 60,000 images and test has 10,000. Both sets have the full 784 features. Every pixel value is normalized by being divided by 255. This is due to there being 255 values on the grayscale image. This ensures that every pixel value is within [0,1], which scales the data, improving model convergence and allows for machine learning classification methods that will be used later.

3 Classic Classification Methods

4 baseline models are created to classify MNIST. Firstly a logistic regression is created. This regression is a of a linear combination of the 784 features which is fed through the sigmoid functon which shrinks the values to the range [0,1]. This output represents the probability that the input belongs to the target class.

$$z = w_0 + w_1x_1 + w_2x_2 + \ldots + w_px_p \qquad P(y = 1|x) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

The second classification is a decision tree which captures non-linear relationships. The tree recursively splits the data into smaller, more homogeneous groups, choosing the split that best reduces Gini impurity,

$$Gini(N) = 1 - \Sigma_{i=1}^{K} p_i^2$$

where p_i is the proportion of class samples. In our case, the maximum depth was set to 16, which was the point where test accuracy peaked, training accuracy continued to increase beyond this depth, but test accuracy did not. Allowing greater depths makes the tree overly specific to the training set, creating tiny, highly tailored leaf nodes that fail to generalize, a classic form of overfitting. Therefore, whenever the decision tree is referenced or compared, its depth is assumed to be 16

Finally, we introduce Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA), two generative classifiers that model each class as a Gaussian distribution. LDA assumes all classes share the same covariance matrix, leading to a linear decision boundary, while QDA relaxes this assumption and allows each class to have its own covariance, resulting in curved, more flexible boundaries. Because QDA must estimate more parameters, it can overfit unless we regularize the covariance estimates; in our case, a regularization value of 0.06 gave the best test accuracy before the model became overly flexible. Since the classes in our data have noticeably different spreads, QDA performs better than LDA, and therefore QDA with regularization 0.06 is the version used in the comparisons.

	Test Accuracy	Macro F1 Score	Training Time (s)
Logistic Regression	0.9259	0.9248	118.8428
Decision Tree	0.8833	0.8815	15.9646
QDA	0.9580	0.9579	16.0356

Table 1: Comparison of Classification Methods

Above, Table 1 displays the comparison of the three classification models. Thre three performance features that are being assessed are accuracy, macro f-1 score and training time. Accuracy measures the overall proportion of correct predictions, defined as

$$Accuracy = \frac{TP + TN}{TotalPredictions}$$

Macro F-1 score computes the average F-1 score from each class. Where the F-1 score is a harmonic mean of precision and recall.

$$F1 = 2\frac{Precision\ x\ Recall}{Precision\ +\ Recall} \qquad Macro\ F1 = \frac{1}{K}\Sigma_{i=1}^{K} F1_i$$

Looking at the results it is clear to see QDA is the strongest peformer with the highest test accuracy (0.9580) and macro F-1 score (0.9579). Logistic regression is second with an accuracy of 0.9259 and macro F-1 score of 0.9248 but has by far the longest training time at over 100 seconds. Decision tree is the weakest model, accuracy (0.8833) and macro F-1 score (0.8815) but the quickest training time at 15.96 seconds. Although its training time is fast it cannot capture the distribution of data as efficiently as logistic regression and QDA, Overall the QDA offers the best accuracy to training time trade off. A confusion matrix visualizing the performance of QDA is shown below in Figure 2

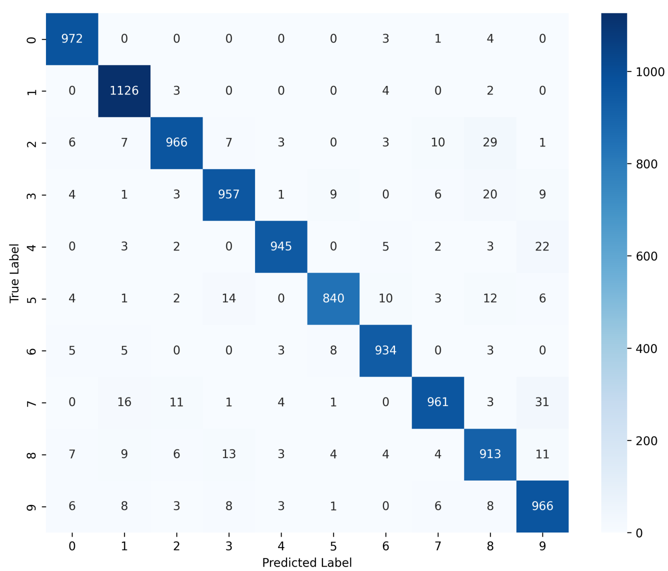

Figure 2: Confusion Matrix of QDA

4 Dimensionality Reduction with PCA

The performances of the classification methods may be improved by implementing principal component analysis (PCA). PCA finds the directions in data where variance is largest and projects the data onto those directions. This reduces the dimensions of data and removes noise whilst preserving most of the variance. The reduction of features can cause training times to reduce, help prevent overfitting and make classed easier to seperate, which could be useful to boost performance in the previous models. In this case the original 784 features will be reduced down to 10,50 and 100 principal components where the previous 3 models will be ran on and compared to see if PCA improves model performance.

The generation of the three PCAs show that 10 components capture 48.8 percent of the variance, 50 captures 82.5 percent and 100 captures 91.5 percent. The logistic regression, decision tree and QDA are all trained and tested on these 3 different PCA dimensions and are compared below in Table 3

	Test Accuracy	Macro F1 Score	Training Time (s)
Logistic Regression (10 PCA)	0.8091	0.8057	2.3860
Decision Tree (10 PCA)	0.8329	0.8305	1.9329
QDA (10 PCA)	0.8907	0.8892	0.0870
Logistic Regression (50 PCA)	0.9117	0.9103	4.5214
Decision Tree (50 PCA)	0.8434	0.8410	10.1421
QDA (50 PCA)	0.9661	0.9659	0.2574
Logistic Regression (100 PCA)	0.9218	0.9206	7.2643
Decision Tree (100 PCA)	0.8401	0.8377	20.2239
QDA (100 PCA)	0.9645	0.9643	0.7022

Table 2: Comparison of Methods with different PCA components

Table 3 summarizes the performance of the three models when the dimensionality of the data is reduced from 784 features to 10, 50, and 100 principal components. For both Logistic Regression and the Decision Tree, a consistent pattern emerges. Test accuracy and macro F1-scores decline relative to the baseline across all levels of dimensionality reduction, although these values progressively approach the original performance as more components are retained. Training times for these models initially decrease substantially, most notably for Logistic Regression, but increase again as the number of components grows, with the Decision Tree at 100 components surpassing its baseline training time. QDA exhibits a different trend. While its training time increases as more components are added, consistent with the behaviour of the other models, its predictive performance improves past its bassline once sufficient variance is preserved. The most notable improvement occurs at 50 principal components, where QDA achieves its highest performance across all experiments, with an accuracy of 0.9661 and a macro F1-score of 0.9659, surpassing both its baseline and all other PCA configurations. Although QDA at 100 components remains strong (accuracy 0.9645, macro F1 0.9643, training time 0.702 s), the peak performance at 50 components indicates that moderate dimensionality reduction provides the optimal balance of information retention and noise suppression for this model.

Overall, the results show that PCA can both help and hinder performance depending on how aggressively the dimensionality is reduced and which classification model is applied. It is shown that the best combination of this is the QDA with 50 principle components, where predictive power and computational efficiency surpass the baseline. The other models lost predictive power form the reduction but increased computational efficiency. Different methods benefit differently from PCA. As PCA removes noise and correlation in the data it benefits the logistic regression and QDA but causes disruption in the decision tree process of effective threshold based splitting.

5 Neural Network

To delve beyond the classic classification and PCA-based models, a deep learning approach is explored using a neural network in an attempt to find a model with better predictive power. Neural networks are able to learn complex non linear patterns within datasets making this a natural next step in a search for finding the best classification model. The network takes the 784 pixel values as inputs, then runs through a series of layers which apply a sequence of weighted transformations followed by nonlinear activation functions. This allows the model to learn increasingly abstract representations of digit shapes. The network produces an output of 10 features, which calculates the probability for each class, then a loss of difference between the predicted and true labels are computed and a weight attached, which is then minimized by an optimizer. To ensure the model generalizes well and is not overfitting to the data, validation accuracy is monitored, in addition to this 5 fold cross validation in performed training the model on 5 different subsets of the training data and averaging the results giving more reliable and less sensitive estimates of the models performance due to the stochastic nature of neural networks.

Neural networks have many possible architectures due to all features of the network being interdependent, interacting in complex ways. A complete search to find the best neural network is computationally infeasible because of the many combinations. As a result, the model is searched by sequential narrowing, creating a baseline network, and then looping through features to find the best validation accuracy until the optimal neural network architecture is identified. A bassline neural network is engineered, which gives a validation accuracy of 0.9806. After sequential narrowing, the best neural network has the following architecture. 2 hidden layers with (256,224) neurons, (ReLu, Tanh) activation functions, Adam optimiser, 0.001 learning rate and a batch size of 256. This neural network has a validation accuracy of 0.9822. There is improvement but modest, the additional neurons increase capacity model non-linear relationships within the image, the most likely cause for the increase in accuracy is the change in activations ReLu has a range of all positive numbers enabling high level learning of features in the first layer whilst tanh has a range of [-1,1] uncovering negative relations which ReLu and sigmoid cannot.

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Expanding on a standard neural network a subset of a neural network is explored, a convolutional neural network (CNN). A CNN flattens images into a one-dimensional vector designed to conserve the spatial structure of the image data then convolutional layers apply filters across local regions of the image, which allow the model to detect patterns of the image efficiently. The main difference between a CNN and NN is the parameter size, due to the parameter sharing of CNN using the same filter everywhere, there are significantly less parameters that need to be explored a CNN compared to a NN. Now that both deep learning methods have been constructed, they are run upt to 30 iterations to check training/validation accuracy and loss to detect overfitting. This is shown in Figure 3.

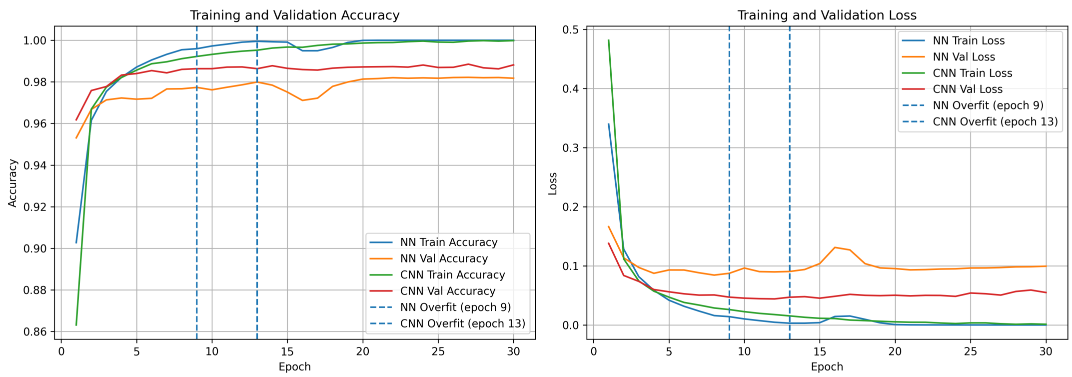

Figure 3: Training Curves for NN and CNN

Figure 3 shows rapid increase in accuracy and decrease in loss for both models showing effective training. Overfitting happens when the training and validation curves diverge from each other. This occurs mildly at epoch 9 for NN and less pronounced at 14 for CNN. This indicates that the CNN has better generalization as it maintains more stable validation accuracy and lower loss for a longer period. The NN and CNNs are now trained and tested at the given epochs where their performance metrics are now compared to the best classic classification method, QDA, shown in Table 3

	Test Accuracy	Macro F1 Score	Training Time (s)
QDA	0.9661	0.9659	0.2574
Neural Network	0.9822	0.982	32.618
CNN	0.9912	0.991	26.135

Table 3: QDA v NN v CNN Performance

Table 3 shows that the NN is has significantly higher test accuracy and macro F1 score than the QDA but has significantly higher training time, taking around 130 times longer. This is due to the increased complexity of neural networks and the iterative processes of large numbers of features. The CNN strongly dominates the NN with improved accuracy and training time reaching 99+ percent accuracy, However it still takes 100 times as long as the QDA. There exists a clear trade off between predictive power and computational efficiency. More accurate methods require more time for the model to train and process the data compared to least accurate ones.

6 Final Analysis

From all of the classification models created the best for deployment of predicting images form the MNIST dataset is the convolutional neural network with test accuracy of 0.9912, macro F1 score of 0.9900 and training time of 26.135

seconds. However this may not be the case for every classification problem. If the main priority of the classifying was speed the best model would be the QDA with a time of 0.2574 seconds whilst still maintaining roughly 96 percent accuracy. However, in the case of identifying handwritten numbers the applications of identifying these can be of high importance for example cheques and credit card information where it is vital for the numbers to be classified correctly therefore for this classification a CNN is best for deployment.

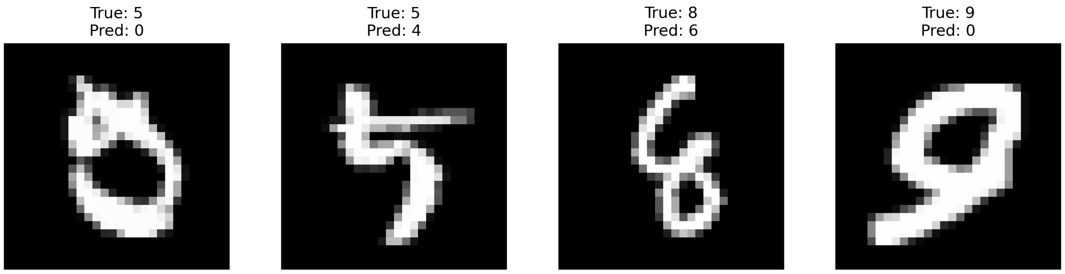

Figure 4: Wrong Predictions with NN

Figure 4 displays a couple of predictions that the CNN got wrong, the numbers are drawn very poorly and would even be mistaken by humans. It could be argued that the values look more like the predicted than the true value for example the first image is missing the pattern of the straight line at the top and the curve at the bottom nearly connect all the way round into a circle which is a pattern 0 would have. This shows the patterns that the CNN follows, straight lines and curves of white pixels at certain points on the image.

This leads on to a limitation of the classification methods, the MNIST dataset has only 70,000 low resolution, greyscale images with some human error providing ambiguous images. The low resolution and greyscale images simplifies the classification task which inflate performances across all models and constrasting human errors can bias model pattern recognition skewing results. MNIST does not necessarily reflect how the classification models would perform on more complex, colored and larger real life image scenarios. The second limitation is that most of these models that were explored were sensitive to model architecture here fine tuning of hyperparameters were essential to performance. The looping of possible architectures were very computationally extensive and even infeasible in deep learning models, this provides a limit of exploration depending on CPU and GPU capabilities and potential performance metrics.

7 Add if can

- Finish conclusion
- Read over everything make sure its explains all the discuss options
- Try add visualisation of LDA QDA
- Fine tune stuff to add/fix